Chase Bank – TAP #25070 Crossroads and 12th

Geotechnical Engineering Report

Rochester, MN

October 10, 2025 | Terracon Project No. JY255038

Prepared for:

The Architects Partnership, Ltd.
200 South Michigan Avenue, Suite 1020
Chicago, IL 60604

Prepared by:

Terracon Consultants

I hereby certify that this plan, specification, or report was prepared by me or under my direct supervision and that I am a duly licensed Professional Engineer under the laws of the state of Minnesota.

Kyle Shubert, P.E.
October 10, 2025
License Number: 42675

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October 10, 2025

The Architects Partnership, Ltd.
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Attn: Terron Wright
P: (561) 628-9845
E: wright@tapchicago.com

Re: Geotechnical Engineering Report
Chase Bank – TAP #25070 Crossroads and 12th
Crossroads Dr. SW and U.S. Hwy 14
Rochester, MN
Terracon Project No. JY255038

Dear Mr. Wright:

We have completed the scope of Geotechnical Engineering services for the above referenced project in general accordance with Terracon Proposal No. PJY255038 dated June 24, 2025. This report presents the findings of the subsurface exploration and provides geotechnical recommendations concerning earthwork and the design and construction of foundations, floor slabs, and pavements for the proposed project.

We appreciate the opportunity to be of service to you on this project. If you have any questions concerning this report or if we may be of further service, please contact us.

Sincerely,

Terracon

Elton Musa

Elton Musa
Senior Staff Engineer

Kyle Shubert, P.E.
Operations Manager

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Note: This report was originally delivered in a web-based format. **Blue Bold** text in the report indicates a referenced section heading. The PDF version also includes hyperlinks which direct the reader to that section and clicking on the Terracon logo will bring you back to this page. For more interactive features, please view your project online at **client.terracon.com**.

Refer to each individual Attachment for a listing of contents.

Introduction

This report presents the results of our subsurface exploration and Geotechnical Engineering services performed for the proposed Chase Bank to be located near the intersection of Crossroads Dr. SW and U.S. Hwy 14 in Rochester, MN. The purpose of these services was to provide information and geotechnical engineering recommendations relative to:

- Subsurface soil conditions
- Groundwater conditions
- Site preparation and earthwork
- Foundation design and construction
- Floor slab design and construction
- Pavement design and construction
- Frost considerations

The geotechnical engineering Scope of Services for this project included the advancement of test borings, laboratory testing, engineering analysis, and preparation of this report.

Drawings showing the site and boring locations are shown on the **Site Location** and **Exploration Plan**, respectively. The results of the laboratory testing performed on soil samples obtained from the site during our field exploration are included on the boring logs in the **Exploration and Laboratory Results** section.

Project Description

Our initial understanding of the project was provided in our proposal and was discussed during project planning. A period of collaboration has transpired since the project was initiated, and our final understanding of the project conditions is as follows:

Item	Description
Information Provided	A Request for Proposal (RFP) was received via email from Terron Wright of The Architects Partnership, Ltd. on June 19, 2025. The request included the following documents: ▪ 20250603_Rochester Crossroads_Test Fit_DRAFT.pdf ▪ Crossroads and 12th Olmsted County GIS.pdf ▪ Rochester crossroads General Development Plan.pdf

Item	Description
Project Description	The project includes the construction of a new bank building at the southwest corner of the intersection of Crossroads Drive SW and U.S. Highway 14 in Rochester, Minnesota.
Proposed Structure	We understand that the project will include the construction of a single story, slab-on-grade (non-basement) bank building with a detached trash enclosure.
Building Construction	Not provided; we anticipate construction of the bank building will be cast in place concrete or concrete masonry unit foundations with steel framing and the trash enclosure will be a concrete masonry unit structure.
Finished Floor Elevation	Not provided; we assume that final grades will be within 2 feet of existing grades.
Maximum Loads	Not provided; in the absence of information provided by the design team, we will use the following loads in estimating settlement based on our experience with similar projects. ▪ Columns: 75 kips ▪ Walls: 5 kips per linear foot (klf) ▪ Slabs: 150 pounds per square foot (psf)
Grading/Slopes	Not provided; we assume that final grades will be within 2 feet of existing grades. Final slopes should not be steeper than 3:1 (Horizontal: Vertical).
Below-Grade Structures	None anticipated.
Free-Standing Retaining Walls	Retaining walls are not expected to be constructed as part of site development to achieve final grades.
Pavements	We assume both rigid (concrete) and flexible (asphalt) pavement sections are being considered. We assume traffic loading will not exceed 50,000 ESALs over a pavement design life of 20 years.

Terracon should be notified if any of the above information is inconsistent with the planned construction, especially the grading limits, as modifications to our recommendations may be necessary.

Site Conditions

The following description of site conditions is derived from our site visit in association with the field exploration and our review of publicly available geologic and topographic maps.

Item	Description
Parcel Information	The project is located at the southwest corner of the intersection of Crossroads Drive SW and U.S. Highway 14 in Rochester, Minnesota. Approximate Coordinates: Latitude: 44.006576° N, Longitude: -92.468893° W See **Site Location**.
Existing Improvements	The property is vacant and grass covered. Review of historical aerial photographs and the information provided by Mr. Wright indicates the property was formerly occupied by an apartment building.
Current Ground Cover	The site is covered with grass.
Existing Topography	Based on our review of MnTOPO, the site appears to be relatively flat with an elevation of about 1004 feet.

Geotechnical Characterization

We have developed a general characterization of the subsurface conditions based upon our review of the subsurface exploration, laboratory data, geologic setting and our understanding of the project. This characterization, termed GeoModel, forms the basis of our geotechnical calculations and evaluation of the site. Conditions observed at each exploration point are indicated on the individual logs. The individual logs can be found in the **Exploration and Laboratory Results** and the GeoModel can be found in the **Figures** attachment of this report.

As part of our analyses, we identified the following model layers within the subsurface profile. For a more detailed view of the model layer depths at each boring location, refer to the GeoModel.

Model Layer	Layer Name	General Description
1	**Fill**	Variable composition; clayey sand, silty sand, and poorly graded sand with silt, occasional gravel and debris (wood, glass, Styrofoam, concrete), brown to black in color.
2	**Native Sand**	Sand with varying amounts of silt and gravel (SP, SP-SM, SM), fine to coarse grained, brown in color, with a very loose to medium dense relative density.

We noted existing fill at the boring locations that appeared to extend to depths ranging from about 4 to 8.7 feet. Borings B-3, B-3A, and B-3B obstructed in fill at a depth of

8.7 feet. Based on our review of public historical aerial photos and geologic maps, we anticipated the existing fill is associated with the former development of the property.

Groundwater Conditions

The boreholes were observed while drilling for the presence and level of groundwater. Groundwater was observed in three of the borings at depths of about 17 feet. It should be noted that groundwater levels measured during drilling represent short-term conditions and may not reflect stabilized levels. A relatively long period is often necessary for the groundwater level to develop and stabilize within a borehole. Long-term monitoring through piezometers or observation wells, properly sealed from surface water infiltration, is recommended if precise groundwater level data are required for design purposes.

Groundwater level fluctuations occur due to seasonal variations in the amount of rainfall, runoff and other factors not evident at the time the borings were performed. Therefore, groundwater levels during construction or at other times in the life of the structure will vary. The possibility of groundwater level fluctuations should be considered when developing the design and construction plans for the project.

Geotechnical Overview

The site appears suitable for the proposed construction based upon geotechnical conditions encountered in the test borings, provided that the recommendations provided in this report are implemented in the design and construction phases of this project.

The main geotechnical concern on the subject property is the existing fill soil observed at the boring locations that was previously placed during development of the surrounding properties. Items that may affect development and construction include:

- **Undocumented Existing Fill Soils:** As noted in **Geotechnical Characterization**, existing fill was encountered in the borings performed across the site to depths ranging from about 4 to 8.7 feet below existing grades. Documentation regarding placement and compaction of the existing fill was not provided for our review. The presence of existing fill on the site presents risks that unsuitable materials or voids may be present and concealed by the fill. If existing fill remains in place below planned structures (foundations, slabs, or pavements), there is an elevated risk of unacceptable settlement. Structures supported over undocumented fills often do not perform predictably.

  Based on the lack of documentation of the fill, we consider the existing fill to be undocumented, and not suitable for support of foundations and floor slabs. Our recommendations therefore include removal of the existing fill from below the building structure and the appropriate oversize area and replacement with

structural fill. The **Shallow Foundations** section addresses shallow foundation considerations for support of the structures directly bearing on engineered fill.

Support of pavements on or above existing fill materials is discussed in this report. However, even with the recommended construction procedures, an inherent risk remains for the owner that compressible fill or unsuitable material, within or buried by the fill, will not be discovered. This risk of unforeseen conditions cannot be eliminated without completely removing the existing fill but can be reduced by following the recommendations contained in this report. To take advantage of the cost benefit of not removing the entire amount of undocumented fill, the owner must be willing to accept the risk of increased differential performance which can result in increased cracking and abrupt differential settlement. Should this risk be acceptable, pavements can be supported above the fill as outlined in **Existing Fill** in the **Earthwork** section.

- **Buried Construction Debris**: Debris, including wood, glass, and Styrofoam was observed within the undocumented existing fill soil in Boring B-3. Boring B-3 also obstructed at a depth of about 8.7 feet on what appeared to be concrete. Borings offset to the north and east of B-3 also obstructed at depths of about 8.7 feet. Debris, including footings, slabs, and utilities associated with prior development should be removed from the building structure footprint and oversize area and properly disposed off-site.

  Please note that reuse of the existing fill soil may require sorting debris-laden soils or removing debris from the soils prior to reuse. Sorting of excavated material can add significant time and expense to a project.

The recommendations contained in this report are based upon the results of field and laboratory testing (presented in the **Exploration and Laboratory Results**), engineering analyses, and our current understanding of the proposed project. The **General Comments** section provides an understanding of the report limitations.

Earthwork

Earthwork is anticipated to include clearing and grubbing, excavations, and engineered fill placement. The following sections provide recommendations for use in the preparation of specifications for the work. Recommendations include critical quality criteria, as necessary, to render the site in the state considered in our geotechnical engineering evaluation for foundations, floor slabs, and pavements.

Site Preparation

The following table includes anticipated excavation depths for removal of the existing fill from the building structures, based on borings completed to evaluate foundation options

for the proposed structures. Please note that actual excavation depths of the unsuitable soils will vary and may be deeper than represented in the table.

Boring	Approximate Surface Elevation (feet MSL)	Depth of Unsuitable Soil (feet)	Anticipated Excavation Elevation Bottom (feet MSL)
B-1	1004	6	998
B-2	1004	4	1000
B-3, B-3A, B-3B	1004	> 8.7 [1]	--
B-4	1004	4	1000
B-5	1003	4	999

1. Borings B-3, B-3A, and B-3B obstructed at a depth of about 8.7 feet on apparent concrete. Removal of unsuitable soil and construction debris is anticipated to extend below a depth of 8.7 feet.

At a minimum in pavement areas, the surficial organic portions of the existing fill should be removed. Support of pavements on the existing fill is discussed in the **Existing Fill** section.

Subgrade Preparation

After removal of unsuitable soils, the exposed areas which will receive fill, once properly cleared and benched where necessary, should be scarified to a minimum depth of 12 inches, moisture conditioned as necessary, and compacted per the compaction requirements in this report. Compacted structural fill soils should then be placed to the proposed design grade and the moisture content and compaction of subgrade soils should be maintained until foundation or pavement construction.

Based upon the subsurface conditions determined from the geotechnical exploration, subgrade soils exposed during construction are anticipated to be relatively workable; however, the workability of the subgrade may be affected by precipitation, repetitive construction traffic or other factors. If unworkable conditions develop, workability may be improved by scarifying and drying.

Existing Fill

As noted in **Geotechnical Characterization**, the borings encountered previously placed fill to depths ranging from about 4 to 8.7 feet. We have no records to indicate the degree of control, and consequently, the fill is considered unreliable for support of foundation and floor slab loads. Support of pavements on or above existing fill soils is discussed in this report. However, even with the recommended construction procedures,

inherent risk exists for the owner that compressible fill or unsuitable material, within or buried by the fill, will not be discovered. This risk of unforeseen conditions cannot be eliminated without completely removing the existing fill but can be reduced by following the recommendations contained in this report.

If the pavements will be supported on the non-organic existing fill, the following protocol should be followed. Once the planned subgrade elevation has been reached, the entire pavement area should be scarified to a minimum depth of 12 inches, moisture conditioned as necessary, and surface compacted per compaction requirements in this report. Areas of soft or otherwise unsuitable material should be undercut and replaced with either new structural fill or suitable, existing on-site materials.

Excavation

We anticipate that excavations for the proposed construction can be accomplished with conventional earthmoving equipment. The bottom of excavations should be thoroughly cleaned of loose soils and disturbed materials prior to backfill placement and/or construction.

Soil Stabilization

Methods of subgrade improvement, as described below, could include scarification, moisture conditioning and recompaction, or removal of unstable materials and replacement with granular fill (with or without geosynthetics). The appropriate method of improvement, if required, would be dependent on factors such as schedule, weather, the size of area to be stabilized, and the nature of the instability. More detailed recommendations can be provided during construction as the need for subgrade stabilization occurs. Performing site grading operations during warm seasons and dry periods would help reduce the amount of subgrade stabilization required.

If the exposed subgrade is unstable during proofrolling operations, it could be stabilized using one of the methods outlined below.

- **Scarification and Recompaction** - It may be feasible to scarify, dry, and recompact the exposed soils. The success of this procedure would depend primarily upon favorable weather and sufficient time to dry the soils. Stable subgrades likely would not be achievable if the thickness of the unstable soil is greater than about 1 foot, if the unstable soil is at or near groundwater levels, or if construction is performed during a period of wet or cool weather when drying is difficult.
- **Crushed Stone** - The use of crushed stone or crushed gravel is a common procedure to improve subgrade stability. The use of high modulus geotextiles (i.e., engineering fabric or geogrid) could also be considered after underground work such as utility construction is completed. Prior to placing the fabric or geogrid, we

recommend that all below grade construction, such as utility line installation, be completed to avoid damaging the fabric or geogrid. Equipment should not be operated above the fabric or geogrid until one full lift of crushed stone fill is placed above it. The maximum particle size of granular material placed over geotextile fabric or geogrid should not exceed 1-½ inches.

- **Chemical Modification** - Improvement of subgrades with portland cement or class C fly ash could be considered for improving unstable soils. Chemical modification should be performed by a pre-qualified contractor having experience with successfully stabilizing subgrades in the project area on similar sized projects with similar soil conditions. Results of chemical analysis of the additive materials should be provided to the geotechnical engineer prior to use. The hazards of chemicals blowing across the site or onto adjacent property should also be considered. Additional testing would be needed to develop specific recommendations to improve subgrade stability by blending chemicals with the site soils. Additional testing could include, but not be limited to, determining the most suitable stabilizing agent, the optimum amounts required, the presence of sulfates in the soil, and freeze-thaw durability of the subgrade.

Further evaluation of the need and recommendations for subgrade stabilization can be provided during construction as the geotechnical conditions are exposed.

Fill Material Types

Reuse of On-Site Soil: Excavated on-site soils are generally considered to be suitable for reuse as structural fill at this site provided they are free of organic material and debris. Portions of the existing fill materials may require removal of debris prior to reuse. In addition, portions of the on-site soil have an elevated fines content and will be sensitive to moisture conditions (particularly during seasonally wet periods) and may not be suitable for reuse when above optimum moisture content. Moisture conditioning (drying) may be required with on-site fine-grained soils to achieve the recommended levels of compaction.

Imported Fill Materials: Imported fill materials should meet the following material property requirements. Regardless of its source, compacted fill should consist of approved materials that are free of organic matter and debris. Frozen material should not be used, and fill should not be placed on a frozen subgrade.

Soil Type [1]	USCS Classification	Acceptable Parameters (for Structural Fill)
Granular	GP, GP-GM, SP, SP-SM	Maximum particle size of 3 inches Less than 12% passing No. 200 sieve

1. Fill should consist of approved materials free of organic matter and debris. Frozen material should not be used, and fill should not be placed on a frozen subgrade. A sample of each material type should be submitted to the Geotechnical Engineer for evaluation prior to use on this site. Additional geotechnical consultation should be provided prior to use of uniformly graded gravel on the site.

Fill Placement and Compaction Requirements

Structural and general fill should meet the following compaction requirements.

Item	Structural Fill	General Fill
Maximum Lift Thickness	8 inches or less in loose thickness when heavy, self-propelled compaction equipment is used 4 to 6 inches in loose thickness when hand-guided equipment (i.e. jumping jack or plate compactor) is used	Same as structural fill
Minimum Compaction Requirements [1, 2]	98% of maximum dry density below foundations, floor slabs, and pavements	92% of maximum
Water Content Range [1]	Granular: -3% to +3% of optimum	As required to achieve minimum compaction requirements

1. Maximum density and optimum water content as determined by the standard Proctor test (ASTM D698). We recommend structural fill be tested for moisture content and compaction during placement. Should the results of the in-place density tests indicate the specified moisture or compaction limits have not been met, the area represented by the test should be reworked and retested as required until the specified moisture and compaction requirements are achieved.
2. Moisture levels of granular materials should be maintained at levels satisfactory for compaction to be achieved without the coarse-grained fill material bulking during placement or pumping when proofrolled.

Utility Trench Backfill

Any soft or unsuitable materials encountered at the bottom of utility trench excavations should be removed and replaced with structural fill or bedding material in accordance

with public works specifications for the utility to be supported. This recommendation is particularly applicable to utility work requiring grade control and/or in areas where subsequent grade raising could cause settlement in the subgrade supporting the utility. Trench excavation should not be conducted below a downward 1:1 projection from existing foundations without engineering review of shoring requirements and geotechnical observation during construction.

On-site materials are considered suitable for backfill of utility and pipe trenches from 1 foot above the top of the pipe to the final ground surface, provided the material is free of organic matter and deleterious substances.

Trench backfill should be mechanically placed and compacted as discussed earlier in this report. Compaction of initial lifts should be accomplished with hand-operated tampers or other lightweight compactors. Where trenches are placed beneath slabs or footings, the backfill should satisfy the gradation and expansion index requirements of engineered fill discussed in this report. Flooding or jetting for placement and compaction of backfill is not recommended.

Grading and Drainage

All grades must provide effective drainage away from the building during and after construction and should be maintained throughout the life of the structure. Water retained next to the building can result in soil movements greater than those discussed in this report. Greater movements can result in unacceptable differential floor slab and/or foundation movements, cracked slabs and walls, and roof leaks. The roof should have gutters/drains with downspouts that discharge onto splash blocks at a distance of at least 10 feet from the building.

Exposed ground should be sloped and maintained at a minimum 5% away from the building for at least 10 feet beyond the perimeter of the building. Locally, flatter grades may be necessary to transition ADA access requirements for flatwork. After building construction and landscaping have been completed, final grades should be evaluated to document effective drainage has been achieved. Grades around the structure should also be periodically inspected and adjusted, as necessary, as part of the structure's maintenance program. Where paving or flatwork abuts the structure, a maintenance program should be established to effectively seal and maintain joints and prevent surface water infiltration.

Earthwork Construction Considerations

Shallow excavations for the proposed structure are anticipated to be accomplished with conventional construction equipment. Upon completion of filling and grading, care should be taken to maintain the subgrade water content prior to construction of grade-supported improvements such as floor slabs and pavements. Construction traffic over

the completed subgrades should be avoided. The site should also be graded to prevent ponding of surface water on the prepared subgrades or in excavations. Water collecting over or adjacent to construction areas should be removed. If the subgrade freezes, desiccates, saturates, or is disturbed, the affected material should be removed, or the materials should be scarified, moisture conditioned, and recompacted prior to floor slab construction.

As a minimum, excavations should be performed in accordance with OSHA 29 CFR, Part 1926, Subpart P, "Excavations" and its appendices, and in accordance with any applicable local and/or state regulations.

Construction site safety is the sole responsibility of the contractor who controls the means, methods, and sequencing of construction operations. Under no circumstances shall the information provided herein be interpreted to mean Terracon is assuming responsibility for construction site safety or the contractor's activities; such responsibility shall neither be implied nor inferred.

Excavations or other activities resulting in ground disturbance have the potential to affect adjoining properties and structures. Our scope of services does not include review of available final grading information or consider potential temporary grading performed by the contractor for potential effects such as ground movement beyond the project limits. A preconstruction/precondition survey should be conducted to document nearby property/infrastructure prior to any site development activity. Excavation or ground disturbance activities adjacent or near property lines should be monitored or instrumented for potential ground movements that could negatively affect adjoining property and/or structures.

Construction Observation and Testing

The earthwork efforts should be observed by the Geotechnical Engineer (or others under their direction). Observation should include documentation of adequate removal of surficial materials (vegetation, topsoil, and pavements), evaluation and remediation of existing fill materials, as well as proofrolling and mitigation of unsuitable areas delineated by the proofroll.

Each lift of compacted fill should be tested, evaluated, and reworked, as necessary, as recommended by the Geotechnical Engineer prior to placement of additional lifts. Each lift of fill should be tested for density and water content at a frequency of at least one test for every 1,000 square feet of compacted fill in the building areas and 2,500 square feet in pavement areas. Where not specified by local ordinance, one density and water content test should be performed for every 100 linear feet of compacted utility trench backfill and a minimum of one test performed for every 12 vertical inches of compacted backfill.

In areas of foundation excavations, the bearing subgrade should be evaluated by the Geotechnical Engineer. If unanticipated conditions are observed, the Geotechnical Engineer should prescribe mitigation options.

In addition to the documentation of the essential parameters necessary for construction, the continuation of the Geotechnical Engineer into the construction phase of the project provides the continuity to maintain the Geotechnical Engineer's evaluation of subsurface conditions, including assessing variations and associated design changes.

Shallow Foundations

If the site has been prepared in accordance with the requirements noted in **Earthwork**, the following design parameters are applicable for shallow foundations.

Design Parameters – Compressive Loads

Item	Description
Maximum Net Allowable Bearing Pressure [1, 2]	3,000 psf - foundations bearing upon recompacted native sand or granular engineered fill
Required Bearing Stratum [3]	Recompacted native sand or engineered fill extending to recompacted native sand
Minimum Foundation Dimensions	Columns: 30 inches Continuous: 18 inches
Ultimate Passive Resistance [4] **(equivalent fluid pressures)**	360 pcf (granular backfill)
Sliding Resistance [5]	0.30 allowable coefficient of friction (granular material)
Minimum Embedment below Finished Grade [6]	Exterior footings in unheated areas: 60 inches Exterior footings in heated areas: 42 inches Interior footings in heated areas: 12 inches
Estimated Total Settlement from Structural Loads [2]	Less than about 1 inch
Estimated Differential Settlement [2, 7]	About 2/3 of total settlement

1. The maximum net allowable bearing pressure is the pressure in excess of the minimum surrounding overburden pressure at the footing base elevation. Values assume that exterior grades are no steeper than 20% within 10 feet of structure.
2. Values provided are for maximum loads noted in **Project Description**. Additional geotechnical consultation will be necessary if higher loads are anticipated.

3. Unsuitable or soft soils should be overexcavated and replaced per the recommendations presented in **Earthwork**.
4. Use of passive earth pressures require the sides of the excavation for the spread footing foundation to be nearly vertical and the concrete placed neat against these vertical faces or that the footing forms be removed and compacted structural fill be placed against the vertical footing face. Assumes no hydrostatic pressure.
5. Can be used to compute sliding resistance where foundations are placed on suitable soil/materials. Frictional resistance for granular materials is dependent on the bearing pressure which may vary due to load combinations. For fine-grained materials, lateral resistance using cohesion should not exceed ½ the dead load.
6. Embedment necessary to minimize the effects of frost and/or seasonal water content variations. For sloping ground, maintain depth below the lowest adjacent exterior grade within 5 horizontal feet of the structure.
7. Differential settlements are noted for equivalent-loaded foundations and bearing elevation as measured over a span of 50 feet.

Design Parameters – Overturning

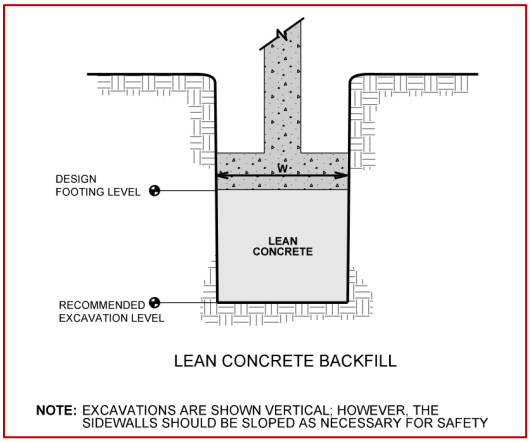
Shallow foundations subjected to overturning loads should be proportioned such that the resultant eccentricity is maintained in the center-third of the foundation (e.g., $e < b/6$, where b is the foundation width). This requirement is intended to keep the entire foundation area in compression during the extreme lateral/overturning load event. Foundation oversizing may be required to satisfy this condition.

Foundation Construction Considerations

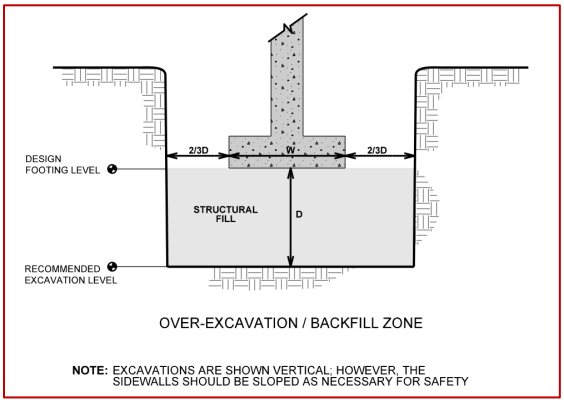
As noted in **Earthwork**, the footing excavations should be evaluated under the observation of the Geotechnical Engineer. The base of all foundation excavations should be free of water and loose soil, prior to placing concrete. Concrete should be placed soon after excavating to reduce bearing soil disturbance. Care should be taken to prevent wetting or drying of the bearing materials during construction. Excessively wet or dry material or any loose/disturbed material in the bottom of the footing excavations should be removed/reconditioned before foundation concrete is placed.

Sensitive soils exposed at the surface of footing excavations may require surficial compaction with hand-held dynamic compaction equipment prior to placing structural fill, steel, and/or concrete. Should surficial compaction not be adequate, construction of a working surface consisting of either crushed stone or a lean concrete mud mat may be required prior to the placement of reinforcing steel and construction of foundations.

If unsuitable bearing soils are observed at the base of the planned footing excavation, the excavation should be extended deeper to suitable soils, and the footings could bear directly on these soils at the lower level or on lean concrete backfill placed in the excavations. The lean concrete replacement zone is illustrated on the sketch below.



Overexcavation for structural fill placement below footings should be conducted as shown below. The overexcavation should be backfilled up to the footing base elevation, with granular structural placed, as recommended in the **Earthwork** section.



Floor Slabs

Item	Description
Floor Slab Support [1]	Minimum 6 inches of free-draining (less than 5% passing the No. 200 sieve) crushed aggregate Subgrade compacted to recommendations in **Earthwork**
Estimated Modulus of Subgrade Reaction [2]	125 pounds per square inch per inch (psi/in) for point loads for slabs

1. Floor slabs should be structurally independent of building footings or walls to reduce the possibility of floor slab cracking caused by differential movements between the slab and foundation.
2. Modulus of subgrade reaction is an estimated value based upon our experience with the subgrade condition, the requirements noted in **Earthwork**, and the floor slab support as noted in this table. It is provided for point loads. For large area loads, the modulus of subgrade reaction would be lower.

The use of a vapor retarder should be considered beneath concrete slabs on grade covered with wood, tile, carpet, or other moisture sensitive or impervious coverings, when the project includes humidity-controlled areas, or when the slab will support equipment sensitive to moisture. When conditions warrant the use of a vapor retarder, the slab designer should refer to ACI 302 and/or ACI 360 for procedures and cautions regarding the use and placement of a vapor retarder.

Saw-cut contraction joints should be placed in the slab to help control the location and extent of cracking. For additional recommendations, refer to the ACI Design Manual. Joints or cracks should be sealed with a waterproof, non-extruding compressible compound specifically recommended for heavy duty concrete pavement and wet environments.

Where floor slabs are tied to perimeter walls or turn-down slabs to meet structural or other construction objectives, our experience indicates differential movement between the walls and slabs will likely be observed in adjacent slab expansion joints or floor slab cracks beyond the length of the structural dowels. The Structural Engineer should account for potential differential settlement through use of sufficient control joints, appropriate reinforcing or other means.

Floor Slab Construction Considerations

Finished subgrade, within and for at least 10 feet beyond the floor slab, should be protected from traffic, rutting, or other disturbance and maintained in a relatively moist condition until floor slabs are constructed. If the subgrade should become damaged or desiccated prior to construction of floor slabs, the affected material should be removed, and structural fill should be added to replace the resulting excavation. Final conditioning of the finished subgrade should be performed immediately prior to placement of the floor slab support course.

The Geotechnical Engineer should observe the condition of the floor slab subgrades immediately prior to placement of the floor slab support course, reinforcing steel, and concrete. Attention should be paid to high traffic areas that were rutted and disturbed earlier, and to areas where backfilled trenches are located.

Pavements

General Pavement Comments

Pavement designs are provided for the traffic conditions and pavement life conditions as noted in **Project Description** and in the following sections of this report. A critical aspect of pavement performance is site preparation. Pavement designs noted in this

section must be applied to the site which has been prepared as recommended in the **Earthwork** section.

Pavement Design Parameters

Pavement thickness can be determined in general accordance with the Minnesota Department of Transportation (MnDOT), which generally follows AASHTO (1993) guidelines, Asphalt Institute and/or other methods if specific wheel loads, axle configurations, frequencies, and desired pavement life are provided. The American Concrete Institute (ACI) ACI 330R-08 – Guide for the Design and Construction of Concrete Parking Lots was utilized to formulate recommended Portland cement concrete pavement sections.

For the parking lot areas, we have assumed pavements would have traffic loads less than 50,000 18-kip Equivalent Single Axle Loads (ESALs) over a 20-year design life.

Pavement Section Thicknesses

Opinions of pavement thicknesses are based on the subsurface conditions encountered at the borings, general characterization of the subgrade, our experience on similar projects, and consider that the subgrade is proofrolled, tested, and evaluated as recommended in this report. Testing such as California Bearing Ratio (CBR), resilient modulus (MR), resistance value (R-Value), etc. was not part of our scope of service for this project to evaluate the support characteristics of the subgrade; however, these can be performed upon request. The thickness of pavements for these scenarios should be in accordance with local city or county ordinances.

The following tables summarizes the estimated minimum Portland cement concrete (PCC) and asphaltic cement concrete (ACC) pavement thicknesses for the anticipated traffic conditions for the facility. These sections are based on the subsurface conditions encountered at the borings and our experience on similar projects and consider that all materials are placed on a subgrade prepared and evaluated as recommended in this report.

Asphaltic Concrete Design

Layer [1]	Thickness (inches)
ACC [2]	4 (two lifts)
Aggregate Base	8

1. Pavement materials, mix design, and construction should conform to the current Minnesota Department of Transportation (MnDOT) Standard Specifications for Construction.
2. A minimum 1.5-inch surface course should be used on ACC pavements.

The following table provides our estimated minimum thickness of PCC pavements.

Portland Cement Concrete Design

Layer [1]	Thickness (inches)
PCC [2]	5
Aggregate Base [3]	6

1. Pavement materials, mix design, and construction should conform to the current Minnesota Department of Transportation (MnDOT) Standard Specifications for Construction.
2. PCC pavement concrete should have a minimum 28-day compressive strength of 4,000 psi.
3. The crushed aggregate base course materials (MnDOT Class 5 MnDOT Spec 3138.2E) should be placed on a stable subgrade which is compacted to at least 98 percent of the material's standard Proctor maximum dry density. Considers the subgrade is sloped to promote drainage.

Thicker pavement sections could be used to reduce maintenance and extend the expected service life of the pavements.

We recommend using a minimum 7-inch thick PCC pavement in areas of anticipated concentrated loads (e.g., loading and unloading areas, dumpster pads) and areas with repeated turning or maneuvering of trucks (e.g., entrance aprons). We also recommend PCC pavement sections include sufficient reinforcing steel and dowels at joints to hold joints tightly closed and to provide load transfer across transverse joints and to reduce differential movement between pavement slabs. Trash container pads should be large enough to support the container and the tipping axle of the collection truck.

Construction traffic on the pavements was not considered in developing the recommended minimum pavement thicknesses. If the pavements will be subject to traffic by construction equipment/vehicles, the pavement thicknesses should be revised to consider the effects of the additional traffic loading.

Although not required for structural support, a minimum 6-inch thick base course layer is recommended to help reduce potential for slab curl, shrinkage cracking, and subgrade pumping through joints. Proper joint spacing will also be required to prevent excessive slab curling and shrinkage cracking. Joints should be sealed to prevent entry of foreign material and doweled where necessary for load transfer. PCC pavement details for joint spacing, joint reinforcement, and joint sealing should be prepared in accordance with ACI 330 and ACI 325.

Where practical, we recommend early-entry cutting of crack-control joints in PCC pavements. Cutting of the concrete in its "green" state typically reduces the potential for micro-cracking of the pavements prior to the crack control joints being formed, compared to cutting the joints after the concrete has fully set. Micro-cracking of

pavements may lead to crack formation in locations other than the sawed joints, and/or reduction of fatigue life of the pavement.

Openings in pavements, such as decorative landscaped areas, are sources for water infiltration into surrounding pavement systems. Water can collect in the islands and migrate into the surrounding subgrade soils thereby degrading support of the pavement. Islands with raised concrete curbs, irrigated foliage, and low permeability near-surface soils are particular areas of concern. The civil design for the pavements with these conditions should include features to restrict or collect and discharge excess water from the islands. Examples of features are edge drains connected to the stormwater collection system, longitudinal subdrains, or other suitable outlets and impermeable barriers preventing lateral migration of water such as a cutoff wall installed to a depth below the pavement structure.

Pavement Drainage

Pavements should be sloped to provide rapid drainage of surface water. Water allowed to pond on or adjacent to the pavements could saturate the subgrade and contribute to premature pavement deterioration. In addition, the pavement subgrade should be graded to provide positive drainage within the granular base section. Appropriate sub-drainage or connection to a suitable daylight outlet should be provided to remove water from the granular subbase.

Due to frost-susceptible soils and the possibility of perched groundwater, consideration should be given to installing a pavement subdrain system to control subgrade moisture, improve stability, and improve long-term pavement performance.

Subsurface drainage systems (i.e., a permeable base and subdrains) below pavement areas generally prolong the life of a pavement and help to prevent infiltrated surface water from becoming trapped below pavements. Saturation of the pavement subgrade could result in a reduction of subgrade strength (rutting) and/or possible heaving. The use of a granular subbase could also be considered to reduce the potential for frost action.

At a minimum, we recommend installing finger drains around catch basins, and drain tile at low points of the pavement subgrade. Finger drains should be installed to drain into catch basins, consisting of four drain lines, spaced at 90 degrees around the catch basin, and extending at least 10 feet out from the catch basin. The pavement subgrade should slope toward the subdrain lines. The subdrains should be hydraulically connected to the free-draining granular base layer (if included in the design). Subdrains should be sloped to provide positive gravity drainage to reliable discharge points. Periodic maintenance of subdrains is required for long-term proper performance.

Pavement Maintenance

The pavement sections represent minimum recommended thicknesses and, as such, periodic upkeep should be anticipated. Preventive maintenance should be planned and provided for through an ongoing pavement management program. Maintenance activities are intended to slow the rate of pavement deterioration and to preserve the pavement investment. Pavement care consists of both localized (e.g., crack and joint sealing and patching) and global maintenance (e.g., surface sealing). Additional engineering consultation is recommended to determine the type and extent of a cost-effective program. Even with periodic maintenance, some movements and related cracking may still occur, and repairs may be required.

Pavement performance is affected by its surroundings. In addition to providing preventive maintenance, the civil engineer should consider the following recommendations in the design and layout of pavements:

- Final grade adjacent to paved areas should slope down from the edges at a minimum 2%.
- Subgrade and pavement surfaces should have a minimum 2% slope to promote proper surface drainage.
- Install pavement drainage systems surrounding areas anticipated for frequent wetting.
- Install joint sealant and seal cracks immediately.
- Seal all landscaped areas in or adjacent to pavements to reduce moisture migration to subgrade soils.
- Place compacted, low permeability backfill against the exterior side of curb and gutter.

Frost Considerations

The clayey sand and silty sand soils on this site are frost susceptible, and small amounts of water can affect the performance of the slabs on-grade, sidewalks, and pavements. Exterior slabs should be anticipated to heave during winter months. If frost action needs to be eliminated in critical areas, we recommend the use of non-frost susceptible (NFS) fill or structural slabs (for instance, structural stoops in front of building doors). Placement of NFS material in large areas may not be feasible; however, the following recommendations are provided to help reduce potential frost heave:

- Provide surface drainage away from the building and slabs, and toward the site drainage system.
- Install drains around the perimeter of the building, stoops, below exterior slabs and pavements, and connect them to the site drainage system.

- Grade clayey subgrades so groundwater potentially perched in overlying fill or aggregate base, slope toward a site drainage system.
- Place NFS fill as backfill beneath slabs and pavements critical to the project.
- Place a 3 horizontal to 1 vertical (3H:1V) transition zone between NFS fill and other soils.
- Place a 3H:1V transition zone of NFS fill adjacent to structural stoops.
- Place NFS materials in critical sidewalk areas.

General Comments

Our analysis and opinions are based upon our understanding of the project, the geotechnical conditions in the area, and the data obtained from our site exploration. Variations will occur between exploration point locations or due to the modifying effects of construction or weather. The nature and extent of such variations may not become evident until during or after construction. Terracon should be retained as the Geotechnical Engineer, where noted in this report, to provide observation and testing services during pertinent construction phases. If variations appear, we can provide further evaluation and supplemental recommendations. If variations are noted in the absence of our observation and testing services on-site, we should be immediately notified so that we can provide evaluation and supplemental recommendations.

Our Scope of Services does not include either specifically or by implication any environmental or biological (e.g., mold, fungi, bacteria) assessment of the site or identification or prevention of pollutants, hazardous materials or conditions. If the owner is concerned about the potential for such contamination or pollution, other studies should be undertaken.

Our services and any correspondence are intended for the sole benefit and exclusive use of our client for specific application to the project discussed and are accomplished in accordance with generally accepted geotechnical engineering practices with no third-party beneficiaries intended. Any third-party access to services or correspondence is solely for information purposes to support the services provided by Terracon to our client. Reliance upon the services and any work product is limited to our client and is not intended for third parties. Any use or reliance of the provided information by third parties is done solely at their own risk. No warranties, either express or implied, are intended or made.

Site characteristics as provided are for design purposes and not to estimate excavation cost. Any use of our report in that regard is done at the sole risk of the excavating cost estimator as there may be variations on the site that are not apparent in the data that could significantly affect excavation cost. Any parties charged with estimating excavation costs should seek their own site characterization for specific purposes to obtain the specific level of detail necessary for costing. Site safety and cost estimating including

excavation support and dewatering requirements/design are the responsibility of others. Construction and site development have the potential to affect adjacent properties. Such impacts can include damages due to vibration, modification of groundwater/surface water flow during construction, foundation movement due to undermining or subsidence from excavation, as well as noise or air quality concerns. Evaluation of these items on nearby properties are commonly associated with contractor means and methods and are not addressed in this report. The owner and contractor should consider a preconstruction/precondition survey of surrounding development. If changes in the nature, design, or location of the project are planned, our conclusions and recommendations shall not be considered valid unless we review the changes and either verify or modify our conclusions in writing.

Figures

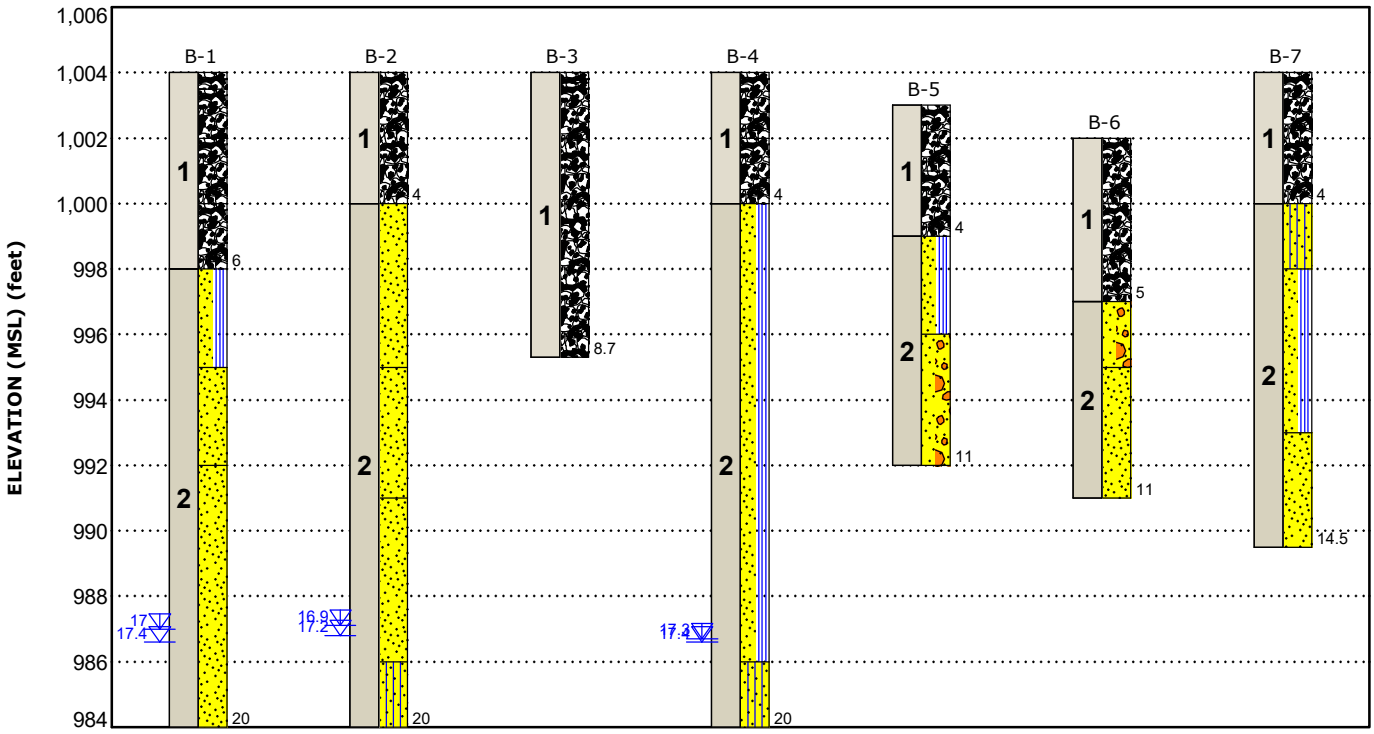
Contents:

GeoModel

Chase Bank - Tap #25070, Crossroads and 12th
Crossroads Drive and 12th Street | Rochester, MN
Terracon Project No. JY255038

Terracon
2625 Highway 14 W, Ste F
Rochester, MN

GeoModel



This is not a cross section. This is intended to display the Geotechnical Model only. See individual logs for more detailed conditions.

Model Layer	Layer Name	General Description
1	Fill	Variable composition; clayey sand, silty sand, and poorly graded sand with silt, occasional gravel and debris (wood, glass, Styrofoam, concrete), brown to black in color.
2	Native Sand	Sand with varying amounts of silt and gravel (SP, SP-SM, SM)), fine to coarse grained, brown in color, with a very loose to medium dense relative density.

Legend

- Fill
- Poorly-graded Sand with Silt
- Poorly-graded Sand
- Silty Sand
- Poorly-graded Sand with Gravel

First Water Observation
Second Water Observation

Groundwater levels are temporal. The levels shown are representative of the date and time of our exploration. Significant changes are possible over time.
Water levels shown are as measured during and/or after drilling. In some cases, boring advancement methods mask the presence/absence of groundwater. See individual logs for details.

NOTES:
Layering shown on this figure has been developed by the geotechnical engineer for purposes of modeling the subsurface conditions as required for the subsequent geotechnical engineering for this project.
Numbers adjacent to soil column indicate depth below ground surface.

Attachments

Exploration and Testing Procedures

Field Exploration

Number of Borings	Approximate Boring Depth (feet)	Location
4	20	Bank Building Corners.[1]
2	11 and 14.5	Paved Parking Areas
1	11	Trash Enclosure

1. Boring B-3, performed in the building pad, encountered auger refusal at a depth of approximately 8.7 feet. To further evaluate subsurface conditions, an offset boring was advanced about 5 feet east of B-3, which also resulted in auger refusal at approximately the same depth (8.7 feet). A second offset boring was then performed approximately 8 feet north and 5 feet east of B-3, which likewise encountered auger refusal at a depth of about 8.7 feet.

Boring Layout and Elevations: Terracon personnel provided the boring layout using handheld GPS equipment (estimated horizontal accuracy of about ±10 feet) and referencing existing site features. Approximate ground surface elevations were estimated from Olmsted County, MN GIS data using the Rochester 2017 elevation contours. If elevations and a more precise boring layout are desired, we recommend borings be surveyed.

Subsurface Exploration Procedures: We advanced the borings with a track-mounted rotary drill rig using continuous flight augers (hollow stem). Five samples were obtained in the upper 11 feet of each boring and at intervals of 5 feet thereafter. In the split-barrel sampling procedure, a standard 2-inch outer diameter split-barrel sampling spoon was driven into the ground by a 140-pound automatic hammer falling a distance of 30 inches. The number of blows required to advance the sampling spoon the last 12 inches of a normal 18-inch penetration is recorded as the Standard Penetration Test (SPT) resistance value. The SPT resistance values, also referred to as N-values, are indicated on the boring logs at the test depths. We observed and recorded groundwater levels during drilling and sampling. For safety purposes, all borings were backfilled with bentonite grout and/or auger cuttings after their completion.

We also observed the boreholes while drilling and at the completion of drilling for the presence of groundwater.

The sampling depths, penetration resistances, and other sampling information were recorded on the field boring logs. The samples were placed in appropriate containers and taken to our soil laboratory for testing and classification by a Geotechnical Engineer. Our exploration team prepared field boring logs as part of the drilling operations. These field

logs included visual classifications of the materials observed during drilling and our interpretation of the subsurface conditions between samples. Final boring logs were prepared from the field logs. The final boring logs represent the Geotechnical Engineer's interpretation of the field logs and include modifications based on observations and tests of the samples in our laboratory.

Laboratory Testing

The project engineer reviewed the field data and assigned laboratory tests. The laboratory testing program included the following types of tests:

- Moisture Content
- Grain Size Analysis (Percent Passing the No. 200 Sieve)

The laboratory testing program often included examination of soil samples by an engineer. Based on the results of our field and laboratory programs, we described and classified the soil samples in general accordance with the Unified Soil Classification System.

Site Location and Exploration Plans

Contents:

Site Location Plan
Exploration Plan

Note: All attachments are one page unless noted above.

Site Location

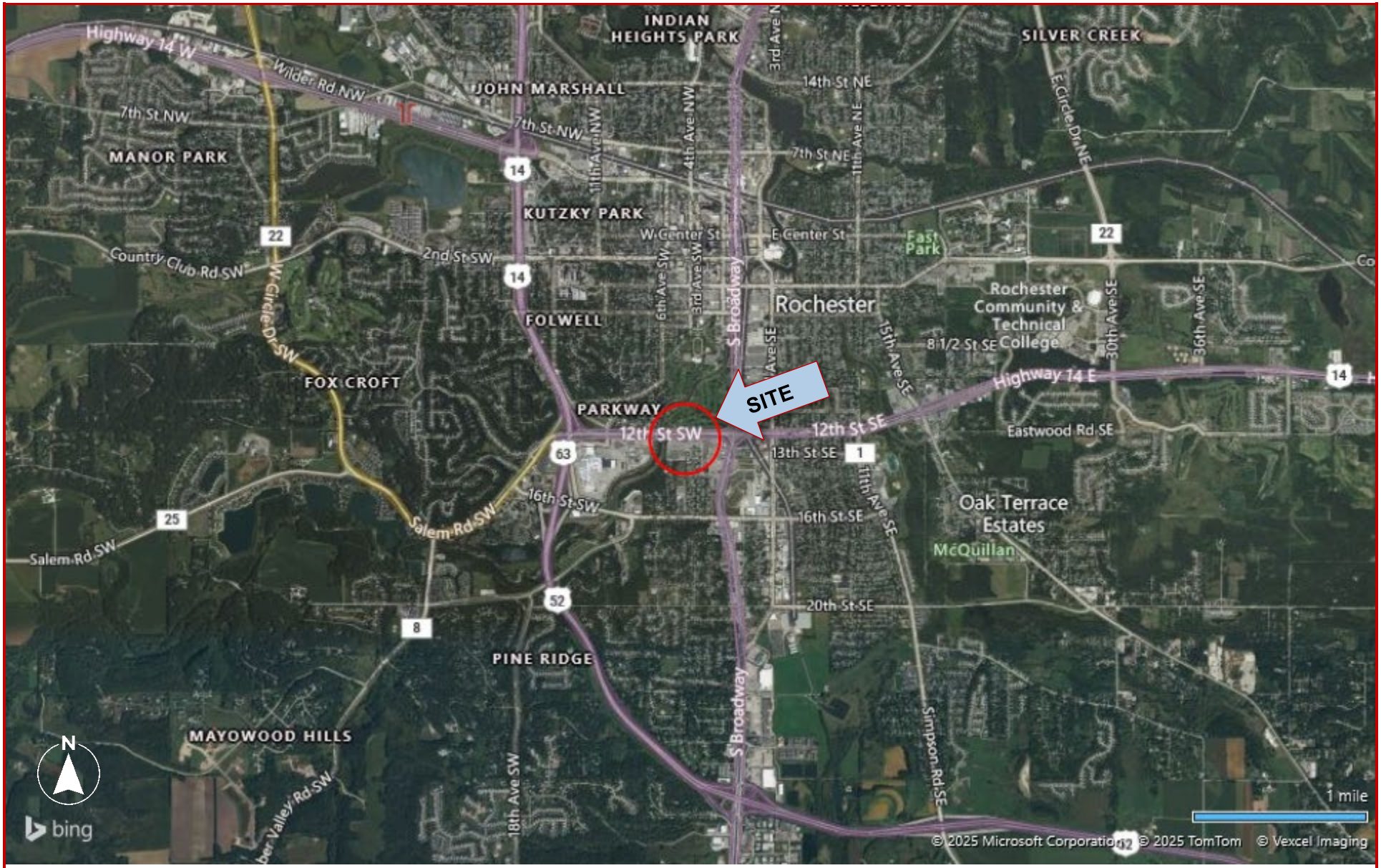


DIAGRAM IS FOR GENERAL LOCATION ONLY, AND IS NOT INTENDED FOR CONSTRUCTION PURPOSES

MAP PROVIDED BY MICROSOFT BING MAPS

Exploration Plan

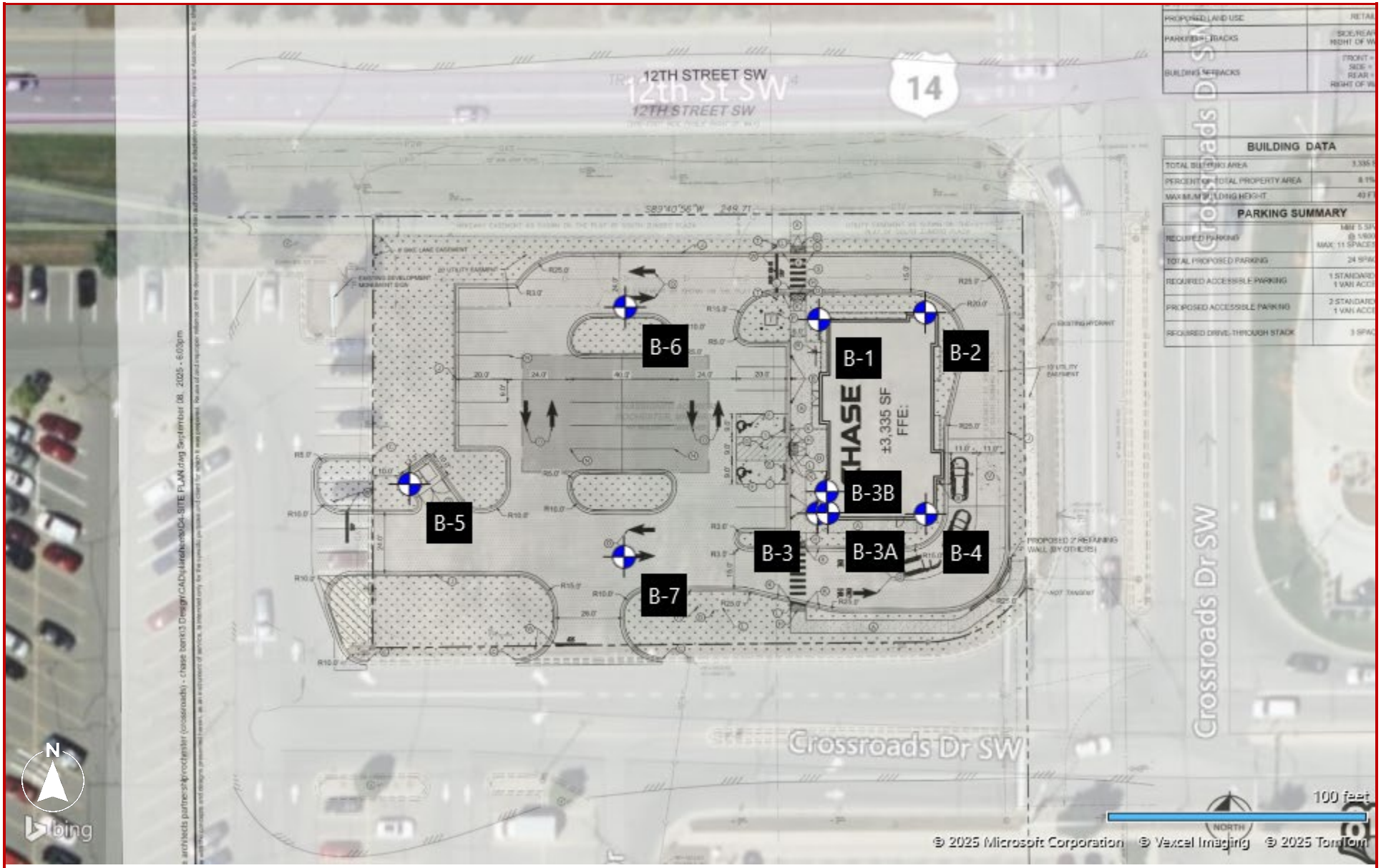


DIAGRAM IS FOR GENERAL LOCATION ONLY, AND IS NOT INTENDED FOR CONSTRUCTION PURPOSES

MAP PROVIDED BY MICROSOFT BING MAPS

Exploration and Laboratory Results

Contents:

Boring Logs (B-1 through B-7)

Note: All attachments are one page unless noted above.

Terracon
2625 Highway 14 W, Ste F
Rochester, MN

Boring Log No. B-1

Location: See Exploration Plan

Latitude: 44.0067° Longitude: -92.4687°

Elevation: 1004 (Ft.)

Model Layer	Depth (Ft.)	Description	Elevation	Recovery (In.)	Field Test Results	Water Content (%)
1	0.5	**FILL - CLAYEY SAND**, trace gravel, fine to medium grained, black	1003.5			
	2.0	**FILL - SILTY SAND**, trace gravel, fine to medium grained, dark brown and brown	1002	7	1-4-6-5 N=10	3.0
	4.0	**FILL - CLAYEY SAND**, trace gravel, fine to medium grained, dark brown and brown	1000	10	1-2-3 N=5	11.7
	6.0	**FILL -POORLY GRADED SAND WITH SILT**, trace gravel, organic fibers, fine to medium grained, brown	998	11	1-2-2 N=4	7.5
2	9.0	**POORLY GRADED SAND WITH SILT (SP-SM)**, trace gravel, fine to coarse grained, brown, very loose	995	9	1-2-2 N=4	6.1
	12.0	**POORLY GRADED SAND (SP)**, trace gravel, fine to coarse grained, brown, loose	992	10	4-3-4 N=7	6.3
	20.0	**POORLY GRADED SAND (SP)**, fine to medium grained, brown, medium dense	984	14	2-7-9 N=16	3.9
				16	2-5-6 N=11	18.0
		Boring Terminated at 20 Feet				

Depth (Ft.): 5, 10, 15, 20

Water Level Observations

Sample Type

Percent Fines

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Elevation Reference: Estimated from MnTOPO Website.

Water Level Observations

Water observed at 17.4 feet while drilling.

Water observed at 17 feet 10 minutes after completion of drilling.

Drill Rig
B-51, DR #1271

Hammer Type
Automatic

Driller
T. Wolfe

Logged by
T. Higgins

Boring Started
09-24-2025

Boring Completed
09-24-2025

Notes

Advancement Method
3 1/4" Hollow Stem Auger

Abandonment Method
Boring backfilled with cuttings from ground surface to 2 feet; backfilled with bentonite grout from 2 feet to boring termination.

Terracon
2625 Highway 14 W, Ste F
Rochester, MN

Boring Log No. B-2

Location: See Exploration Plan

Latitude: 44.0067° Longitude: -92.4685°

Depth (Ft.) Elevation: 1004 (Ft.)

Model Layer	Graphic Log	Description	Depth	Elevation	Depth (Ft.)	Water Level Observations	Sample Type	Recovery (In.)	Field Test Results	Water Content (%)	Percent Fines
1		**FILL - CLAYEY SAND**, trace gravel, fine to medium grained, black and dark brown	4.0	1000				8	2-3-5-8 N=8	9.9	
								9	5-6-6 N=12	12.0	
2		**POORLY GRADED SAND (SP)**, fine to medium grained, brown, medium dense to loose	9.0	995	5			13	5-5-5 N=10	4.0	3.6
								11	2-3-2 N=5	3.8	
		POORLY GRADED SAND (SP), trace gravel, fine to coarse grained, brown, loose	13.0	991	10			12	1-3-5 N=8	5.0	4.0
		POORLY GRADED SAND (SP), fine to medium grained, brown, loose	18.0	986	15			9	1-3-4 N=7	11.7	
		SILTY SAND (SM), trace gravel, silt lenses, fine to medium grained, brown, medium dense	20.0	984	20			14	3-7-9 N=16	20.0	
		Boring Terminated at 20 Feet									

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Elevation Reference: Estimated from MnTOPO Website.

Water Level Observations

Water observed at 17.2 feet while drilling.

Water observed at 16.9 feet 10 minutes after completion of drilling.

Drill Rig
B-51, DR #1271

Hammer Type
Automatic

Driller
T. Wolfe

Logged by
T. Higgins

Boring Started
09-24-2025

Boring Completed
09-24-2025

Notes

Advancement Method
3 1/4" Hollow Stem Auger

Abandonment Method
Boring backfilled with cuttings from ground surface to 2 feet; backfilled with bentonite grout from 2 feet to boring termination.

Terracon
2625 Highway 14 W, Ste F
Rochester, MN

Boring Log No. B-3

Location: See Exploration Plan

Latitude: 44.0065° Longitude: -92.4687°

Depth (Ft.) Elevation: 1004 (Ft.)

Model Layer	Graphic Log	Description	Depth (Ft.)	Water Level Observations	Sample Type	Recovery (In.)	Field Test Results	Water Content (%)	Percent Fines
1		0.3 **FILL - CLAYEY SAND**, fine to medium grained, black 1003.7							
		FILL - POORLY GRADED SAND WITH SILT, trace gravel, with wood pieces, glass and styrofoam, fine to medium grained, black and dark brown				3	1-4-5-3 N=9	12.5	
						5	2-2-2 N=4	4.9	11.8
			5			4	1-1-1 N=2	5.1	
						2	1-1-1 N=2	6.3	
		8.7 Concrete @ 8.7 feet 995.3							
		Auger Refusal at 8.7 Feet							

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Elevation Reference: Estimated from MnTOPO Website.

Water Level Observations
Water not observed while drilling.

Drill Rig
B-51, DR #1271

Hammer Type
Automatic

Driller
T. Wolfe

Notes

Advancement Method
3 1/4" Hollow Stem Auger

Logged by
T. Higgins

Boring Started
09-24-2025

Abandonment Method
Boring backfilled with cuttings from ground surface to 2 feet; backfilled with bentonite grout from 2 feet to boring termination.

Boring Completed
09-24-2025

Terracon
2625 Highway 14 W, Ste F
Rochester, MN

Boring Log No. B-3A

Model Layer	Graphic Log	Location: Offset 5' E of B-3. Latitude: 44.0065° Longitude: -92.4686° Depth (Ft.) Elevation: 1004 (Ft.)	Depth (Ft.)	Water Level Observations	Sample Type	Recovery (In.)	Field Test Results	Water Content (%)	Percent Fines
		Blind drill	5						
		8.7 995.3							
		Auger Refusal at 8.7 Feet							

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).
See Supporting Information for explanation of symbols and abbreviations.
Elevation Reference: Estimated from MnTOPO Website.

Water Level Observations

Drill Rig
B-51, DR #1271

Driller
T. Wolfe

Notes

Advancement Method

Logged by
T. Higgins

Boring Started
09-24-2025

Abandonment Method

Boring Completed
09-24-2025

Terracon
2625 Highway 14 W, Ste F
Rochester, MN

Boring Log No. B-3B

Location: Offset 8' N and 5' E of B-3.

Latitude: 44.0065° Longitude: -92.4686°

Model Layer	Graphic Log	Depth (Ft.)	Elevation: 1004 (Ft.)	Depth (Ft.)	Water Level Observations	Sample Type	Recovery (In.)	Field Test Results	Water Content (%)	Percent Fines
		Blind drill		5						
		8.7	995.3							
		Auger Refusal at 8.7 Feet								

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Elevation Reference: Estimated from MnTOPO Website.

Water Level Observations

Drill Rig
B-51, DR #1271

Driller

Notes

Advancement Method

Logged by

Boring Started
09-24-2025

Abandonment Method

Boring Completed
09-24-2025

Terracon
2625 Highway 14 W, Ste F
Rochester, MN

Boring Log No. B-4

Location: See Exploration Plan

Latitude: 44.0065° Longitude: -92.4685°

Depth (Ft.) Elevation: 1004 (Ft.)

Model Layer	Depth (Ft.)	Description	Elevation
1	0.0	**FILL - CLAYEY SAND**, trace gravel, fine to medium grained, black and dark brown	1002
1	2.0	**FILL - SILTY SAND**, trace gravel, fine to medium grained, brown	1000
2	4.0	**POORLY GRADED SAND WITH SILT (SP-SM)**, trace gravel, fine to medium grained, brown, very loose to medium dense	986
2	18.0	**SILTY SAND (SM)**, trace gravel, fine to medium grained, brown, medium dense	984
	20.0	***Boring Terminated at 20 Feet***	

Depth (Ft.)	Sample Type	Recovery (In.)	Field Test Results	Water Content (%)	Percent Fines
		17	2-3-4-3 N=7	10.4	
		11	2-4-4 N=8	6.5	
5		8	2-1-2 N=3	3.6	
		12	1-2-3 N=5	5.4	5.6
10		14	3-8-12 N=20	3.5	
		15	7-9-11 N=20	3.8	
15					
20		17	5-8-14 N=22	20.5	

Water Level Observations: Water observed at 17.4 feet while drilling; Water observed at 17.3 feet 10 minutes after completion of drilling.

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Elevation Reference: Estimated from MnTOPO Website.

Water Level Observations
Water observed at 17.4 feet while drilling.
Water observed at 17.3 feet 10 minutes after completion of drilling.

Drill Rig
B-51, DR #1271

Hammer Type
Automatic

Driller
T. Wolfe

Logged by
T. Higgins

Boring Started
09-24-2025

Boring Completed
09-24-2025

Notes

Advancement Method
3 1/4" Hollow Stem Auger

Abandonment Method
Boring backfilled with cuttings from ground surface to 2 feet; backfilled with bentonite grout from 2 feet to boring termination.

Terracon
2625 Highway 14 W, Ste F
Rochester, MN

Boring Log No. B-5

Location: See Exploration Plan

Latitude: 44.0065° Longitude: -92.4693°

Depth (Ft.) Elevation: 1003 (Ft.)

Model Layer	Graphic Log	Description	Depth (Ft.)	Elevation	Water Level Observations	Sample Type	Recovery (In.)	Field Test Results	Water Content (%)	Percent Fines
1		**FILL - CLAYEY SAND**, trace gravel, fine to medium grained, dark brown	2.0	1001			7	1-5-7-10 N=12	6.4	
		FILL - SILTY SAND, fine to medium grained, dark brown and brown	4.0	999			13	3-7-10 N=17	9.0	28.7
2		**POORLY GRADED SAND WITH SILT (SP-SM)**, trace gravel, fine to medium grained, brown, medium dense	7.0	996			15	3-8-7 N=15	5.1	
		POORLY GRADED SAND WITH GRAVEL (SP), fine to coarse grained, brown, very loose to medium dense	11.0	992			8	0-1-1 N=2	8.3	
							10	1-6-7 N=13	5.0	
		Boring Terminated at 11 Feet								

Depth (Ft.): 5, 10

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Elevation Reference: Estimated from MnTOPO Website.

Water Level Observations
Water not observed while drilling.

Drill Rig
B-51, DR #1271

Hammer Type
Automatic

Driller
T. Wolfe

Notes

Advancement Method
3 1/4" Hollow Stem Auger

Logged by
T. Higgins

Boring Started
09-24-2025

Abandonment Method
Boring backfilled with cuttings from ground surface to 2 feet; backfilled with bentonite grout from 2 feet to boring termination.

Boring Completed
09-24-2025

Terracon
2625 Highway 14 W, Ste F
Rochester, MN

Boring Log No. B-6

Location: See Exploration Plan

Latitude: 44.0067° Longitude: -92.4689°

Elevation: 1002 (Ft.)

Model Layer	Graphic Log	Depth (Ft.)	Description	Elevation	Depth (Ft.)	Water Level Observations	Sample Type	Recovery (In.)	Field Test Results	Water Content (%)	Percent Fines
1		0.0	**FILL - SILTY SAND**, fine to medium grained, dark brown	1000				10	1-4-5-5 N=9	6.1	
		2.0	**FILL - POORLY GRADED SAND WITH SILT**, fine to medium grained, brown	997				10	1-2-2 N=4	6.6	10.4
2		5.0	**POORLY GRADED SAND WITH GRAVEL (SP)**, fine to coarse grained, brown, loose	995	5			8	1-3-6 N=9	7.8	
		7.0	**POORLY GRADED SAND (SP)**, trace gravel, fine to coarse grained, brown, medium dense to loose	991				14	2-5-7 N=12	5.1	
		11.0	***Boring Terminated at 11 Feet***		10			11	2-1-4 N=5	5.9	

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Elevation Reference: Estimated from MnTOPO Website.

Water Level Observations
Water not observed while drilling.

Drill Rig
B-51, DR #1271

Hammer Type
Automatic

Driller
T. Wolfe

Notes

Advancement Method
3 1/4" Hollow Stem Auger

Logged by
T. Higgins

Boring Started
09-24-2025

Abandonment Method
Boring backfilled with cuttings from ground surface to 2 feet; backfilled with bentonite grout from 2 feet to boring termination.

Boring Completed
09-24-2025

Terracon
2625 Highway 14 W, Ste F
Rochester, MN

Boring Log No. B-7

Location: See Exploration Plan

Latitude: 44.0064° Longitude: -92.4689°

Depth (Ft.) Elevation: 1004 (Ft.)

Model Layer	Graphic Log	Description	Depth (Ft.)	Elevation	Water Level Observations	Sample Type	Recovery (In.)	Field Test Results	Water Content (%)	Percent Fines
1		**FILL - CLAYEY SAND**, trace gravel, fine to medium grained, black and dark brown					15	2-3-3-4 N=6	13.2	
							0	0-0-0 N=0		
			4.0	1000						
2		**SILTY SAND (SM)**, fine to medium grained, brown, very loose	5				13	1-2-2 N=4	7.4	
			6.0	998						
		POORLY GRADED SAND WITH SILT (SP-SM), trace gravel, fine to medium grained, brown, loose to very loose					10	1-3-2 N=5	9.7	
			10				9	3-1-2 N=3	9.3	
			11.0	993						
		POORLY GRADED SAND (SP), trace gravel, fine to medium grained, brown, medium dense					11	2-8-12 N=20	3.7	
			14.5	989.5						
		Boring Terminated at 14.5 Feet								

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Elevation Reference: Estimated from MnTOPO Website.

Water Level Observations
Water not observed while drilling.

Drill Rig
B-51, DR #1271

Hammer Type
Automatic

Driller
T. Wolfe

Notes

Advancement Method
3 1/4" Hollow Stem Auger

Logged by
T. Higgins

Boring Started
09-24-2025

Abandonment Method
Boring backfilled with cuttings from ground surface to 2 feet; backfilled with bentonite grout from 2 feet to boring termination.

Boring Completed
09-24-2025

Supporting Information

Contents:

General Notes
Unified Soil Classification System

Note: All attachments are one page unless noted above.

Terracon
2625 Highway 14 W, Ste F
Rochester, MN

General Notes

Sampling	Water Level	Field Tests
Split Spoon	Water Initially Encountered Water Level After a Specified Period of Time Water Level After a Specified Period of Time Cave In Encountered Water levels indicated on the soil boring logs are the levels measured in the borehole at the times indicated. Groundwater level variations will occur over time. In low permeability soils, accurate determination of groundwater levels is not possible with short term water level observations.	N — Standard Penetration Test Resistance (Blows/Ft.) (HP) — Hand Penetrometer (T) — Torvane (DCP) — Dynamic Cone Penetrometer UC — Unconfined Compressive Strength (PID) — Photo-Ionization Detector (OVA) — Organic Vapor Analyzer

Descriptive Soil Classification

Soil classification as noted on the soil boring logs is based Unified Soil Classification System. Where sufficient laboratory data exist to classify the soils consistent with ASTM D2487 "Classification of Soils for Engineering Purposes" this procedure is used. ASTM D2488 "Description and Identification of Soils (Visual-Manual Procedure)" is also used to classify the soils, particularly where insufficient laboratory data exist to classify the soils in accordance with ASTM D2487. In addition to USCS classification, coarse grained soils are classified on the basis of their in-place relative density, and fine-grained soils are classified on the basis of their consistency. See "Strength Terms" table below for details. The ASTM standards noted above are for reference to methodology in general. In some cases, variations to methods are applied as a result of local practice or professional judgment.

Location And Elevation Notes

Exploration point locations as shown on the Exploration Plan and as noted on the soil boring logs in the form of Latitude and Longitude are approximate. See Exploration and Testing Procedures in the report for the methods used to locate the exploration points for this project. Surface elevation data annotated with +/- indicates that no actual topographical survey was conducted to confirm the surface elevation. Instead, the surface elevation was approximately determined from topographic maps of the area.

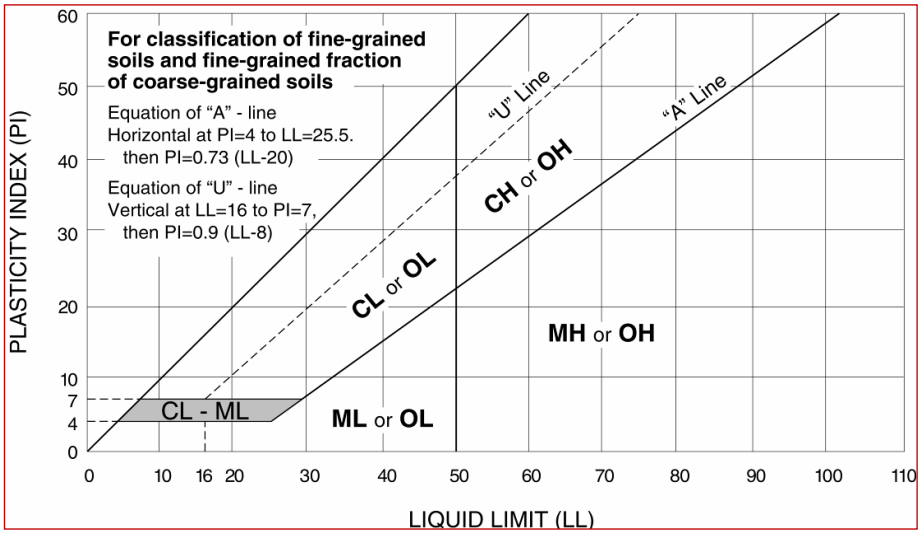
Strength Terms

Relative Density of Coarse-Grained Soils (More than 50% retained on No. 200 sieve.) Density determined by Standard Penetration Resistance		Consistency of Fine-Grained Soils (50% or more passing the No. 200 sieve.) Consistency determined by laboratory shear strength testing, field visual-manual procedures or standard penetration resistance		
Relative Density	**Standard Penetration or N-Value (Blows/Ft.)**	**Consistency**	**Unconfined Compressive Strength Qu (tsf)**	**Standard Penetration or N-Value (Blows/Ft.)**
Very Loose	0 - 3	Very Soft	less than 0.25	0 - 1
Loose	4 - 9	Soft	0.25 to 0.50	2 - 4
Medium Dense	10 - 29	Medium Stiff	0.50 to 1.00	5 - 8
Dense	30 - 50	Stiff	1.00 to 2.00	9 - 15
Very Dense	> 50	Very Stiff	2.00 to 4.00	16 - 30
		Hard	> 4.00	> 30

Relevance of Exploration and Laboratory Test Results

Exploration/field results and/or laboratory test data contained within this document are intended for application to the project as described in this document. Use of such exploration/field results and/or laboratory test data should not be used independently of this document.

Unified Soil Classification System

Criteria for Assigning Group Symbols and Group Names Using Laboratory Tests [A]				Soil Classification	
				Group Symbol	Group Name [B]
Coarse-Grained Soils: More than 50% retained on No. 200 sieve	**Gravels:** More than 50% of coarse fraction retained on No. 4 sieve	**Clean Gravels:** Less than 5% fines [C]	Cu≥4 and 1≤Cc≤3 [E]	GW	Well-graded gravel [F]
			Cu<4 and/or [Cc<1 or Cc>3.0] [E]	GP	Poorly graded gravel [F]
		Gravels with Fines: More than 12% fines [C]	Fines classify as ML or MH	GM	Silty gravel [F, G, H]
			Fines classify as CL or CH	GC	Clayey gravel [F, G, H]
	Sands: 50% or more of coarse fraction passes No. 4 sieve	**Clean Sands:** Less than 5% fines [D]	Cu≥6 and 1≤Cc≤3 [E]	SW	Well-graded sand [I]
			Cu<6 and/or [Cc<1 or Cc>3.0] [E]	SP	Poorly graded sand [I]
		Sands with Fines: More than 12% fines [D]	Fines classify as ML or MH	SM	Silty sand [G, H, I]
			Fines classify as CL or CH	SC	Clayey sand [G, H, I]
Fine-Grained Soils: 50% or more passes the No. 200 sieve	**Silts and Clays:** Liquid limit less than 50	**Inorganic:**	PI > 7 and plots above "A" line [J]	CL	Lean clay [K, L, M]
			PI < 4 or plots below "A" line [J]	ML	Silt [K, L, M]
		Organic:	$\frac{LL\ oven\ dried}{LL\ not\ dried} < 0.75$	OL	Organic clay [K, L, M, N]
					Organic silt [K, L, M, O]
	Silts and Clays: Liquid limit 50 or more	**Inorganic:**	PI plots on or above "A" line	CH	Fat clay [K, L, M]
			PI plots below "A" line	MH	Elastic silt [K, L, M]
		Organic:	$\frac{LL\ oven\ dried}{LL\ not\ dried} < 0.75$	OH	Organic clay [K, L, M, P]
					Organic silt [K, L, M, Q]
Highly organic soils:	Primarily organic matter, dark in color, and organic odor			PT	Peat

[A] Based on the material passing the 3-inch (75-mm) sieve.
[B] If field sample contained cobbles or boulders, or both, add "with cobbles or boulders, or both" to group name.
[C] Gravels with 5 to 12% fines require dual symbols: GW-GM well-graded gravel with silt, GW-GC well-graded gravel with clay, GP-GM poorly graded gravel with silt, GP-GC poorly graded gravel with clay.
[D] Sands with 5 to 12% fines require dual symbols: SW-SM well-graded sand with silt, SW-SC well-graded sand with clay, SP-SM poorly graded sand with silt, SP-SC poorly graded sand with clay.
[E] $Cu = D_{60}/D_{10}$ $\quad Cc = \frac{(D_{30})^2}{D_{10} \times D_{60}}$
[F] If soil contains ≥ 15% sand, add "with sand" to group name.
[G] If fines classify as CL-ML, use dual symbol GC-GM, or SC-SM.
[H] If fines are organic, add "with organic fines" to group name.
[I] If soil contains ≥ 15% gravel, add "with gravel" to group name.
[J] If Atterberg limits plot in shaded area, soil is a CL-ML, silty clay.
[K] If soil contains 15 to 29% plus No. 200, add "with sand" or "with gravel," whichever is predominant.
[L] If soil contains ≥ 30% plus No. 200 predominantly sand, add "sandy" to group name.
[M] If soil contains ≥ 30% plus No. 200, predominantly gravel, add "gravelly" to group name.
[N] PI ≥ 4 and plots on or above "A" line.
[O] PI < 4 or plots below "A" line.
[P] PI plots on or above "A" line.
[Q] PI plots below "A" line.